

Steel grating plate

1 Scope

This section specifies the terminology and definitions of steel grating plates, product structure, model and marking, allowable dimensional deviations, technical requirements, application design, testing methods and inspection rules, ordering content, weight, area, delivery settlement, packaging, marking, and quality certificates.

This section applies to steel grating plates (hereinafter referred to as "steel grating plates") used in work platforms, floors, walkways, stair treads, and gutter covers in industries such as petroleum, chemical engineering, metallurgy, light industry, shipbuilding, energy, and municipal services. Relevant provisions of this section may also be referenced for other types of steel grating plates and metal grating plates.

2 Normative reference documents

The following documents are essential for the application of this document. For dated references, only the version dated applies to this document. For undated references, their latest version (including all change orders) applies to this document.

GB/T 700 Carbon Structural Steel

GB/T 702 Dimensions, Shape, Weight and Allowable Deviations of Hot Rolled Steel Bars

GB/T 709 Dimensions, Shape, Weight and Allowable Deviations of Hot Rolled Steel Plates and Steel Strips

GB/T 1591 Low-alloy high-strength structural steel

GB/T 3280 Stainless Steel Cold-Rolled Plates and Steel Strips

GB/T 4237 Stainless Steel Hot Rolled Plates and Steel Strips

GB/T 13912 Technical Requirements and Test Methods for Hot-Dip Galvanized Metal Coatings on Steel Components

GB 50009 Code for Structural Loads of Buildings

GB 50017 Steel Structure Design Standard

3 Terminology and Definitions

The following terms and definitions apply to this document.

3.1

steel bar grating

A plate-type steel structure component formed by fixing flat steel plates and crossbars arranged at specific intervals in an orthogonal configuration to create open through-holes. Depending on manufacturing methods, it is classified into press-welded steel grating plates and press-locking steel grating plates.

3.2

bearing steel plate

Flat steel bars subjected to loads in steel grating.

3.3

Cross bar

Twisted square steel, round steel, or flat steel fixed on a load-bearing flat steel.

3.4

net space gap

The maximum inscribed circle diameter of open perforations in steel grating.

3.5

central spacing of flat steel bars

The distance between the centers of two adjacent load-bearing flat steel bars.

3.6

Center spacing of cross bars

The distance between the centers of two adjacent horizontal bars.

3.7

Pressure resistance welding

The method of welding by applying pressure through electrodes to workpiece assemblies and utilizing the resistance heat generated by current flowing through the contact surface and adjacent areas of the joint is termed pressure resistance welding, abbreviated as PRW.

3.8

bordure band

Flat steel, angle steel, C-shaped steel, crossbars, or other materials are used as edge plates to secure the ends, sides, or other openings and notches of the flat steel carried by the steel grating.

3.9

Decorative trim band

Steel grating edge plates designed to modify appearance, which neither bear nor transfer loads from flat steel members.

3.10

load-carrying band

At the critical sections of the platform, the load-bearing flat steel ends lack support and rely on the steel grating edge plate with flange plates to bear and transfer the load of the flat steel.

3.11

toe plate

A baffle fixed around the platform or at the edges of steel grating cuts and openings, positioned above the load-bearing flat steel and serving a protective function.

3.12

Steel grid plate length

Maximum size of steel grating parallel to the direction of load-bearing flat steel.

3.13

Steel grid plate width

Maximum size of steel grating perpendicular to the direction of load-bearing flat steel.

3.14

Steel grid plate thickness

The thickness of the steel grid plate equals the width of the load-bearing flat steel.

3.15

span of steel lattice plate

The clear distance of the steel lattice support structure parallel to the direction of bearing flat steel.

3.16

Grating design value of load

The load value of steel grid plate is obtained by calculating the design value of bending strength of steel.

3.17

Design value of concentrated load at mid-span for steel lattice plates

The design value of concentrated load acting on the midline of the steel grid span.

3.18

Design value of steel grid plate under full-span uniform load

The design value of uniformly distributed load acting on the plane area of steel grid plates.

3.19

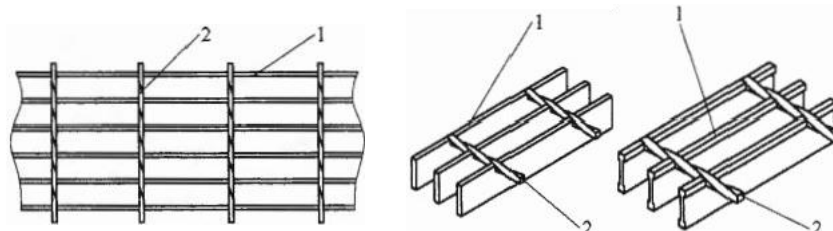
Maximum allowable deflection under steel grid plate load

The maximum allowable deflection at mid-span of steel grating under external loading.

4 Product Structure

4.1 Pressure-welded steel grating

At each intersection point where flat steel and crossbars are supported, the steel grating fixed by pressure resistance welding is referred to as a pressure-welded steel grating. The crossbars of pressure-welded steel gratings are typically made of twisted square steel or round steel, as shown in the figure.



explain :

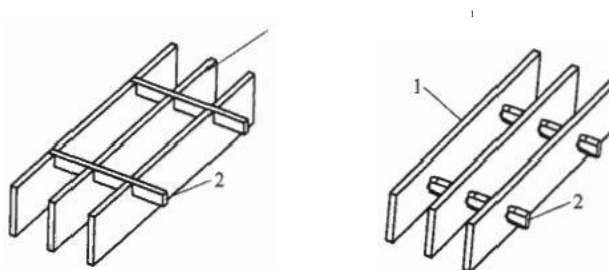
1—Supporting flat steel;

2—Crossbar.

Figure 1 Pressure-welded steel grating

4.2 Pressure-locking steel grating plate

At each intersection point where flat steel plates and crossbars are connected, the crossbars are pressed into pre-cut grooves in the load-bearing flat steel plates using an interference fit assembly method under pressure, or deformed through expansion or forging processes to secure them. Such steel grating plates are referred to as press-lock steel grating plates, as shown in Figure 2.



explain :

1—Supporting

flat steel; 2—

Crossbar.

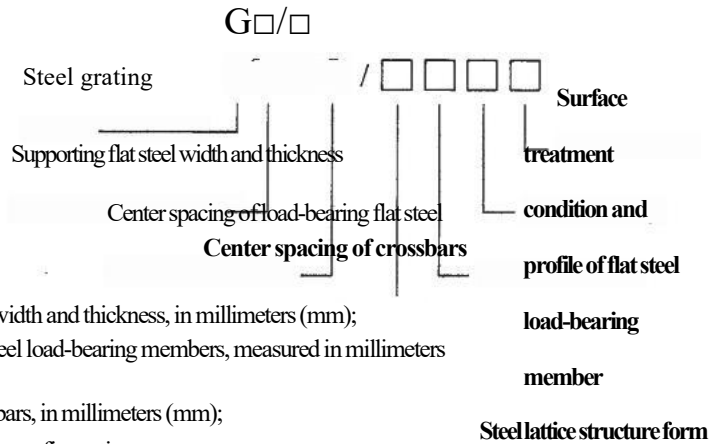
Figure 2 Pressure-locking steel grating plate

4.3 Steel grating panels can be customized to various sizes and shapes according to the requirements of the client.

4.4 Steel grating plates that are not suitable for manufacturing through pressure resistance welding may be produced using alternative methods.

Type 5 and Marking

5.1 Steel grating models are classified into various specifications based on load-bearing flat steel dimensions, spacing between load-bearing flat steel and crossbars, structural configuration of steel grating, cross-sectional shape of load-bearing flat steel, and surface treatment conditions. The model designation method is as follows:



explain :

1. Load-bearing flat steel width and thickness, in millimeters (mm);
2. Center spacing of flat steel load-bearing members, measured in millimeters (mm);
3. Center spacing of crossbars, in millimeters (mm);
4. Steel grating structure configuration:
W — Pressed steel grating (may be omitted in markings);
P — Pressure-locking steel grating.
5. Load-bearing flat steel section shape:
F — flat steel with rectangular cross-section (may be omitted in markings);
Type I steel;
S — Tooth profile flat steel.
6. Surface treatment status:
G — Hot-dip galvanizing (may be omitted in the marking);
U — No surface treatment is applied.

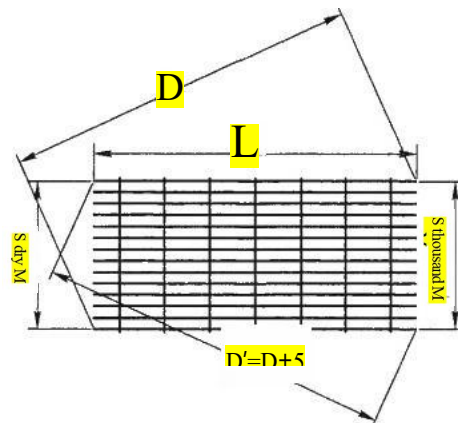
5.2 Labeling Examples

The load-bearing flat steel of the steel grid has a width of 40 mm and a thickness of 5 mm. The center spacing of the load-bearing flat steel is 30 mm, while the center spacing of the crossbars is 100 mm. The structural form is a pressure-welded steel grid with no surface treatment, and its marking is G405/30/100U.

6 Dimensional Allowable Deviation

6.1 The allowable deviation for steel grid plate length is $\pm$ mm, and for width is $\pm$ 5mm, as shown in Figure 3.

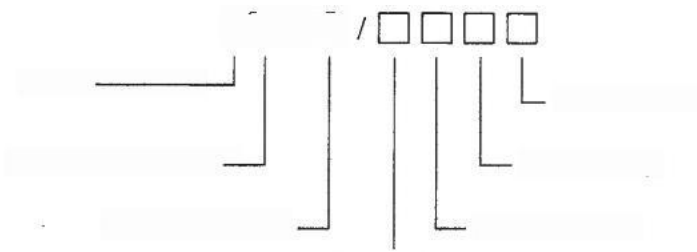
Unit: millimeters (mm)



explain :

- L-length of steel grating;
- W-Steel grating width;
- D — Diagonal length of rectangular steel grating.**

Figure 3 Allowable deviations for length, width, and diagonal dimensions of steel grid plates



6.2 The allowable diagonal deviation for rectangular steel grating plates shall not exceed $\pm 5\text{mm}$, as shown in Figure 3.

6.3 The non-verticality of flat steel load-bearing members shall not exceed 10% of the flat steel width, with the maximum deviation of the lower edge not exceeding 3mm, as shown in Figure 4.

Unit: millimeters (mm)

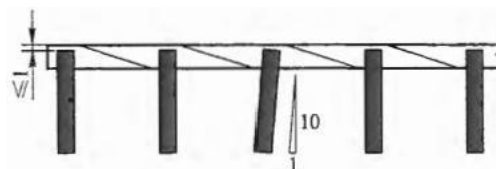


Figure 4 Deviation of flat steel and crossbar bearing

6.4 Crossbar Position Deviation

The surface of the crossbar shall not exceed the surface of the load-bearing flat steel by more than 1 mm, as shown in Figure 4. The ends of the crossbar shall not extend beyond the side faces of the steel grating by more than 2 mm, as shown in Figure 5.

6.5 Crossbar Deflection and Spacing Deviation

The deviation of the crossbar edge from the center of the steel grid plate should not exceed 5mm. Within any 1500mm length, the allowable deviation of the distance between the two ends of the crossbar is $\pm 6\text{mm}$, as shown in Figure 5.

Unit: millimeters (mm)

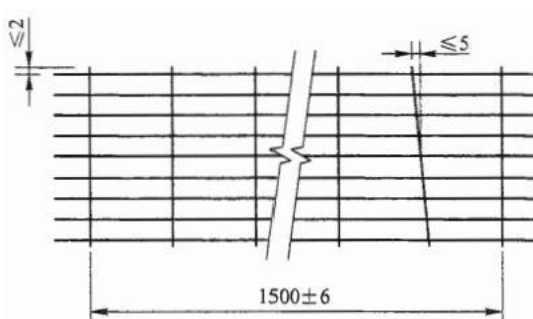
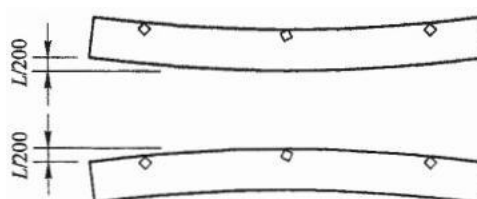


Figure 5 Spacing of crossbars and vertical deviation

6.6 Longitudinal Bending

The longitudinal bending deflection of steel grating shall not exceed $1/200$ of its length, as shown in Figure 6.



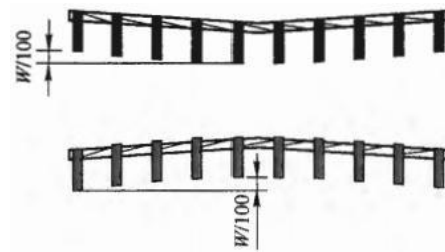
explain :

L-Length of steel grating.

Figure 6 Longitudinal bending

6.7 Transverse Bending

The lateral bending deflection of steel grating shall be less than $1/100$ of its width, as shown in Figure 7.



explain :

W-Width of steel grid plate.

Figure 7 Lateral bending

7 Technical Requirements

7.1 Materials

7.1.1 Bearing flat steel

7.1.1.1 Flat steel load-bearing components may be fabricated from carbon structural steel or low-alloy high-strength structural steel materials, which must comply with the requirements of GB/T 700, GB/T 1591, and GB/T 702; alternatively, austenitic stainless steel or duplex stainless steel materials may be used, provided they meet the specifications of GB/T 4237 and GB/T 3280.

7.1.1.2 The load-bearing flat steel used for steel grid plates shall preferably be hot-rolled or cold-rolled steel strips subjected to longitudinal shear, or alternatively hot-rolled flat steel, I-beams, or other steel sections with specified geometric cross-sections.

7.1.1.3 The spacing between load-bearing flat steel plates and crossbars shall be determined through negotiation between the supplier and the client. For steel grating plates used in work platforms or passageways, the maximum clear spacing between load-bearing flat steel plates shall be less than 35 mm, and the maximum clear spacing between crossbars shall not exceed 150 mm.

7.1.1.4 The load-bearing flat steel bars of steel grating plates may be designed with tooth profiles to enhance the anti-slip performance of the grating. For toothed load-bearing flat steel bars, there shall be no fewer than 5 teeth within any 100mm length segment, with a tooth depth of no less than 1.5mm and a maximum depth not exceeding 2.5mm.

7.1.1.5 The allowable dimensional deviations for flat steel load-bearing members shall comply with the provisions specified in Table 1.

Table 1 Allowable dimensional tolerances for flat steel load-bearing members, in millimeters (mm) Table 1
Allowable dimensional tolerances for flat steel load-bearing members, in millimeters (mm)

Nominal size		allowable deviation
thickness (b)	3 ~ 5	0.3
	>5 ~ 16	Soil 0.4
width (d)	≤50	Student's t 0.4
	>50 ~ 75	0.5
	>75 ~ 150	Version 1.0

7.1.2 Crossbar

7.1.2.1 The crossbar shall preferably be made of the same material as the load-bearing flat steel and must comply with relevant standard requirements.

7.1.2.2 The crossbar shall be made of twisted square steel, round steel, or flat steel, with a cross-sectional area of no less than 20mm².

7.2 Bordering

Welded steel grating should preferably be delivered with edge wrapping. Alternatively, edge wrapping may be omitted upon mutual agreement between the supplier and the purchaser.

7.2.1 Decorative Border

7.2.1.1 Steel grating panels may be provided with decorative edging as required by the client.

7.2.1.2 The decorative edging panel may be fabricated from the same material as the load-bearing flat steel, or alternatively from different metallic or non-metallic materials.

7.2.1.3 The fixing methods for decorative edging panels may include arc welding at intervals of four load-bearing flat steel bars, pressure resistance welding, or other welding techniques, as well as pressing locking, riveting, or bonding. However, the spacing should not exceed 150 mm.

7.2.1.4 Decorative edging: No connection strength test report for decorative edging is required upon delivery.

7.2.2 Load-bearing edging

7.2.2.1 When openings or notches are required at special locations on the platform or for equipment passage, and the load-bearing flat steel lacks support while relying on the flange plate for load bearing and transfer, load-bearing flanges shall be adopted.

7.2.2.2 The load-bearing edge can be welded using flat steel, angle steel, or C-shaped steel, with the connection strength meeting the design load requirements.

7.2.2.3 The load-bearing flange plate shall be welded to each load-bearing flat steel bar without intermittent welding, with at least full single-sided welding. The weld seam shall be an angle butt weld no less than the thickness of the flat steel bar, and the maximum weld height shall not exceed 6 mm. For stair treads and applications subjected to heavy loads, full double-sided welding shall be performed.

7.2.2.4 The welding plan shall be formulated by the platform design department based on actual load design requirements, and the manufacturing plant shall produce according to the design specifications. The additional workload and costs shall be specified in the purchase contract.

7.3 Corrosion Prevention Measures

7.3.1 Steel grating used in industrial corrosive environments should preferably be fabricated from weathering structural steel or stainless steel, with thicker load-bearing flat steel plates selected to enhance corrosion allowance.

7.3.2 Except for stainless steel grating plates, steel grating plates shall preferably undergo hot-dip galvanizing surface treatment.

7.3.3 During steel grating installation, if the anti-corrosion coating is damaged due to welding fixation, cutting, or impact, the on-site construction team and user shall implement remedial anti-corrosion measures.

7.3.4 Stainless steel fasteners may be used for steel grating installation in industrial corrosive environments.

8 Use design

8.1 Installation of Steel Grid Plates

8.1.1 Steel grating shall be installed with its crossbars facing upward on the walkway surface of the platform passage, and must not be installed with crossbars facing downward.

8.1.2 The ends of flat steel beams must be securely supported on the platform's beam structure with reliable fastening; otherwise, personnel access is prohibited.

8.1.3 The dimensions of steel grid plates are constrained by stress during lifting and handling operations. For detachable panels, their size specifications must also account for weight limitations when manually transported.

8.2 Avoid hazards caused by falling objects from gaps in steel grating

8.2.1 The clear spacing of steel grating plates used for work platforms or passageways shall not allow a 35mm -diameter sphere to pass through during descent.

8.2.2 The clear space gaps of steel grating plates used for platforms and passages laid above personnel activity areas, emergency evacuation routes, stair treads, and rest platforms shall not allow a 20mm diameter sphere to pass through upon falling. Otherwise, other appropriate facilities shall be employed to ensure equivalent safety levels.

8.2.3 To prevent hazards caused by objects falling through steel grating plates, the platform and passage steel grating plates must not have gaps exceeding the net spacing of the steel grating plates.

mouth.

8.3 Avoiding pedestrian tripping hazards

8.3.1 To prevent pedestrian tripping, the steel grating platform must be level, with the maximum height difference between adjacent grating panels and between grating panels and surrounding components not exceeding 4mm.

8.3.2 The height difference between the applied load and adjacent unloading steel grating shall not exceed 4mm.

8.4 Risk mitigation for falling steel grating components

8.4.1 The installation and fixation of steel grating plates are specified in Appendix B. To prevent the falling of steel grating components, welding or other fixation methods may be employed during installation.

8.4.2 After installation, the steel grating shall not be moved or detached from the support frame. The support length of the steel grating at both ends in the direction of flat steel load-bearing shall be no less than 25mm per end on the support frame.

8.4.3 All steel grating components shall be securely fastened to the supporting structure and must not rely on adjacent components to prevent lateral movement. Removal of any adjacent component shall not compromise the secure fixation of remaining components.

8.4.4 Avoid using movable detachable steel grating with equal length and width (square). If necessary, additional positioning measures should be implemented to prevent installation errors in the load-bearing direction that may lead to the risk of steel grating collapse.

8.4.5 For components requiring movable and detachable steel grating plates, specialized installation clamps or shot nail fasteners shall be used to secure them, preventing any displacement of the structure. Installation clamps may be supplied by manufacturers as per client specifications. Except for stainless steel clamps, carbon steel clamps and bolts must undergo hot-dip galvanizing or equivalent surface treatment. Bolt diameters shall not be less than 8mm, with no fewer than four clamps required per steel grating plate. Shot nail fasteners shall achieve a penetration depth (measured from nail tip to substrate surface) of at least 10mm while meeting industrial platform design requirements for fastening strength. Fasteners must be selected based on environmental conditions to ensure corrosion resistance. Stainless steel shot nails are mandatory in industrial corrosive environments.

8.4.6 The fastening condition of accessories shall be inspected regularly to detect any corrosion or hazardous loosening, as well as positional changes in clamping components.

8.5 Preventing pedestrian slip hazards

8.5.1 Steel grating plates should exhibit excellent anti-slip performance. For work platforms with slopes not exceeding 10° or environments containing accumulated liquids or oil contaminants, toothed steel grating plates are recommended.

8.5.2 For work platforms with a slope exceeding 10°, more practical anti-slip measures should be implemented to prevent pedestrian slip hazards.

8.6 Dimensions of the Passage for Steel Grid Plate Work Platforms

8.6.1 The width of a single pedestrian passage for steel grating panels shall not be less than 600mm. When the passage is frequently used by pedestrians or involves multiple simultaneous crossings, the width shall be increased to 1200mm.

8.6.2 If steel grating passageways are used as evacuation routes, their width shall comply with specific regulatory requirements. In the absence of such regulations, the width shall not be less than 1600 mm. For high-density crowd evacuation routes, the width of steel grating passageways shall be increased to 2000 mm.

8.7 Design Load for Steel Grid Plate Platform Passage

8.7.1 The design load of steel grid plate work platforms shall comply with the provisions of GB 50009.

8.7.2 Steel grid plate work platforms are typically designed with a uniform load of 2 kN/m². At the most unfavorable position of the platform, a concentrated load of 1.5 kN should be capable of being borne within a 200 mm × 200 mm area.

8.7.3 Maintenance platforms are typically designed to withstand a uniform load of 4 kN/m².

8.7.4 The uniformly distributed load capacity of single-pass steel grating channels shall not be less than 3.0 kN/m².

8.7.5 The steel grating channel with bidirectional traffic shall have a uniform load capacity of no less than 5.0 kN/m².

8.7.6 Steel grating walkways designed for high-density pedestrian traffic must have a uniform load capacity of no less than 7.5 k/m².

8.7.7 When subjected to design loads, the deflection of steel grating plates shall not exceed 1/200 of the span, with a maximum limit of 4 mm.

8.7.8 If notches are present in the steel grating, the remaining area of the grating must meet the design load requirements.

8.7.9 The calculation of the relationship between external loads and deflection for steel grid plates shall be conducted in accordance with the principles specified in GB 50017, as detailed in Appendix C.

8.8 Load Requirements for Steel Grid Plates

8.8.1 The actual deflection under the design value of concentrated load shall not exceed the maximum allowable deflection.

8.8.2 Under the design value of concentrated load, the crossbars of steel grating shall remain welded or locked.

8.8.3 After unloading, the structure must remain stable without deformation, and the residual deflection shall not exceed 1/1000 of the span.

9 Test Methods and Inspection Rules

9.1 Visual inspection of appearance: Each steel grating shall be visually inspected for shape and flatness.

9.2 Dimension Inspection: The dimensions and tolerances of steel grating plates and load-bearing flat steel components must comply with applicable standards and supply contract requirements.

9.3 Load Testing: The manufacturer shall conduct static load testing on steel grating plates in accordance with sampling regulations (see Appendix A for details) and provide test reports as per customer requirements.

9.4 For steel grating with hot-dip galvanized surface treatment, the zinc coating thickness shall comply with the requirements specified in GB/T 13912.

9.5 Product Sampling Regulations

9.5.1 Visual inspection of appearance and dimensional inspection shall be conducted on 10% of each batch, with no fewer than 3 units inspected.

9.5.2 For steel grating with hot-dip galvanized surface treatment, the zinc coating thickness shall be inspected on no fewer than 3 pieces per batch.

9.5.3 Static load testing shall include no fewer than one unit per batch.

9.6 Batch Rules

Steel grating shall be inspected and accepted by batch. Each batch shall consist of steel grating of the same grade, produced using the same forming process and equipment, and of the same model. The quantity of steel grating in each batch shall not exceed the following specifications:

- a) Steel grid plates with thickness $d \leq 30\text{mm}$, quantity: 5,000 pieces;
- b) Steel grid plates with thickness ranging from 30mm to 50mm, quantity: 3,000 pieces;
- c) Steel grid plates with thickness $d > 50\text{mm}$, quantity: 1000 pieces.

10 Order content, weight, area, and delivery settlement

10.1 Order Details for Steel Grid Plates

The contract or order placed under this section shall include the following contents:

- a) Standard number;
- b) Product model specifications;
- c) material ;
- d) Edge finishing requirements and additional workload;
- e) Surface treatment requirements;
- f) Quantity of delivered area;
- g) Theoretical delivery weight;
- h) Number of accompanying accessories;
- i) Additional technical requirements;
- j) Delivery and settlement.

10.2 Both the weight and area of steel grid plates may serve as basis for delivery settlement, subject to mutual agreement between the supplier and the buyer.

10.2.1 Theoretical Weight Calculation of Steel Grid Plates

The theoretical weight of steel grating refers to the weight after edge wrapping and surface treatment (excluding non-surface treatment). The theoretical weight of steel grating is calculated according to Formula (1):

$$W. = (b_1 t_1 N_1 + b_2 t_2 N_2 + 2b_3 t_3) \rho \times 10^{-6} \dots \dots \dots (1)$$

In the formula:

W-Steel grating weight, measured in kilograms per square meter (kg/m^2);

t_1 -width of the flat steel bearing section, in millimeters (mm);

b_1 -the thickness of the flat steel bearing plate, measured in millimeters (mm);

N_1 — Number of flat steel bars per meter of steel grating;

t_2 -Crossbar width, unit: millimeter (mm);

b_2 -Crossbar thickness, unit: millimeter (mm);

N_2 — Number of crossbars per meter of steel grating;

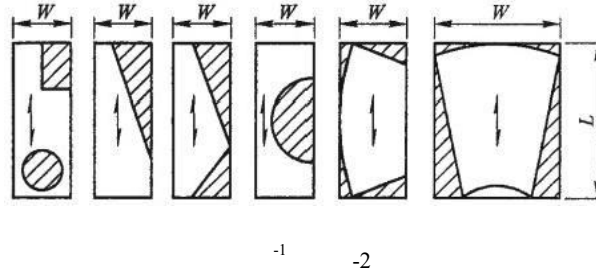
t_3 -Flanged flat steel width, in millimeters (mm);

b_3 -Flanged flat steel thickness, measured in millimeters (mm);

p — Material density, measured in kilograms per cubic meter (kg/m^3);

μ — Surface treatment weight gain coefficient, with hot-dip galvanizing weight gain calculated as 1.06.

10.2.2 The calculated area of irregular steel grating plates with notches is shown in Figure 8. The area is determined based on the total perimeter dimensions on the drawings, including both the opening and notch sections. The area is calculated as width (W) $\times$ length (L).



explain :

1 — Incision;

2—Direction of bearing flat steel.

Figure 8 Irregular steel grating with incision

11 Packaging, Labeling and Quality Certificate

11.1 Packaging

Steel grating is typically manufactured by bundling steel strips, with the weight of each bundle determined through negotiation between the supplier and the buyer or set by the supplier. Upon mutual agreement, the supplier may perform packaging according to the buyer's requirements.

11.2 Markings

The packaging label of steel grating shall indicate the trademark or manufacturer code, steel grating model, and standard number. The steel grating shall be marked with a serial number or traceable code.

11.3 Quality Certificate

The product quality certificate shall specify the product standard number, material grade, model specifications, surface treatment status, appearance and load test report, as well as batch weight. The quality certificate shall be delivered to the user along with the packing list as the basis for acceptance. If requested by the user, a raw material quality assurance certificate shall be provided.

